Lecture 12. Cox Proportional Hazards Regression (Part 2)

Ties, Local Tests, Cut Points, Model Selection, Baseline Hazard

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1 Partial Likelihood with Tied Data

Distinct event times $t_1 < \cdots < t_D$ with $d_i \geq 1$ events at t_i; $D(t_i)$ the set of subjects failing at t_i, $\mathbf{S}_i = \sum_{j \in D(t_i)} \mathbf{Z}_j$, and $R(t_i) = \{j : T_j \geq t_i\}$.

Definition 12.1: Three tie-corrected partial likelihoods

(1) Breslow (1974): ignore the ties.

$$L_1(\boldsymbol{\beta}) = \prod_{i=1}^{D} \prod_{k \in D(t_i)} \frac{\exp(\boldsymbol{\beta}'\mathbf{Z}_k)}{\sum_{j \in R(t_i)} \exp(\boldsymbol{\beta}'\mathbf{Z}_j)} = \prod_{i=1}^{D} \frac{\exp(\boldsymbol{\beta}'\mathbf{S}_i)}{\left[\sum_{j \in R(t_i)} \exp(\boldsymbol{\beta}'\mathbf{Z}_j)\right]^{d_i}}. \tag{8.4.1}$$

(2) Efron (1977):

$$L_2(\boldsymbol{\beta}) = \prod_{i=1}^{D} \frac{\exp(\boldsymbol{\beta}'\mathbf{S}_i)}{\prod_{j=1}^{d_i} \left[\sum_{k \in R(t_i)} \exp(\boldsymbol{\beta}'\mathbf{Z}_k) - \frac{j-1}{d_i} \sum_{k \in D(t_i)} \exp(\boldsymbol{\beta}'\mathbf{Z}_k)\right]}. \tag{8.4.2}$$

(3) Cox (1972), exact (discrete): with Q_i the set of all d_i-subsets $q = \{q_1, \ldots, q_{d_i}\}$ of $R(t_i)$ and $\mathbf{S}_q = \sum_{j=1}^{d_i} \mathbf{Z}_{q_j}$,

$$L_3(\boldsymbol{\beta}) = \prod_{i=1}^{D} \frac{\exp(\boldsymbol{\beta}'\mathbf{S}_i)}{\sum_{q \in Q_i} \exp(\boldsymbol{\beta}'\mathbf{S}_q)}. \tag{8.4.3}$$

- **Breslow**: simple, in most packages; works well with few ties.

- **Efron**: break the ties by $t_i + 0.001, \ldots, t_i + 0.001 d_i$; the tied subjects leave the risk set one by one, each in expectation contributing $1/d_i$ of the tied mass. Closer than Breslow to the exact discrete-model PL; close to Breslow when ties are few.
- **Cox exact**: $h(t \mid \mathbf{Z})$ is the conditional death probability in $(t, t+1)$ given survival to t, and

$$\frac{P(\text{the individuals in } D(t_i) \text{ die at } t_i \mid \text{survival to } t_i)}{P(d_i \text{ deaths at } t_i \mid \text{survival to } t_i)} = \frac{h_0(t_i)\exp(\boldsymbol{\beta}'\mathbf{S}_i)}{\sum_{q \in Q_i} h_0(t_i)\exp(\boldsymbol{\beta}'\mathbf{S}_q)}.$$

  Exact for any d_i, but computationally heavy with many ties.

With no ties all three reduce to (8.3.1); they usually agree unless ties are heavy. Global tests of $H_0 : \boldsymbol{\beta} = \mathbf{0}$ are formed as in Lecture 11.

Example 12.2: catheter placement and exit-site infection

Time to first exit-site infection (months), possibly censored by catheter failure: 43 surgical ($Z = 0$) and 76 percutaneous ($Z = 1$) placements.
Surgical, infections: 1.5, 3.5, 4.5, 4.5, 5.5, 8.5, 8.5, 9.5, 10.5, 11.5, 15.5, 16.5, 18.5, 23.5, 26.5; censored: 2.5, 2.5, 3.5, 3.5, 3.5, 4.5, 5.5, 6.5, 6.5, 7.5, 7.5, 7.5, 7.5, 8.5, 9.5, 10.5, 11.5, 12.5, 12.5, 13.5, 14.5, 14.5, 21.5, 21.5, 22.5, 22.5, 25.5, 27.5.
Percutaneous, infections: 0.5 (×6), 2.5, 2.5, 3.5, 6.5, 15.5; censored: 0.5 (×10), 1.5 (×4), 2.5 (×5), 3.5 (×5), 4.5 (×3), 5.5 (×5), 6.5, 7.5 (×3), 8.5, ...
First event time $t_1 = 0.5$: all 119 subjects are at risk (43 with $Z = 0$, 76 with $Z = 1$), and $d_1 = 6$ infections, all with $Z = 1$, so $S_1 = 6$. The factors at t_1 are

$$\text{Breslow:} \quad \frac{e^{6\beta}}{(43 + 76e^{\beta})^6},$$

$$\text{Efron:} \quad \frac{e^{6\beta}}{\prod_{j=1}^{6}\left[43 + 76e^{\beta} - \frac{j-1}{6}\,6e^{\beta}\right]} = \frac{e^{6\beta}}{\prod_{j=1}^{6}\left[43 + (77-j)e^{\beta}\right]},$$

$$\text{Cox exact:} \quad \frac{e^{6\beta}}{\sum_{k=0}^{6}\binom{43}{6-k}\binom{76}{k}e^{k\beta}},$$

since a 6-subset with k percutaneous members has $S_q = k$.

2 Local Tests

Motivation. Test a subset of the coefficients while the others are left free.

Partition $\boldsymbol{\beta}_{p\times 1} = (\boldsymbol{\beta}_1', \boldsymbol{\beta}_2')'$ with $\boldsymbol{\beta}_1$ of dimension q, test $H_0 : \boldsymbol{\beta}_1 = \mathbf{b}$, and partition

$$I = \begin{pmatrix} I_{11} & I_{12} \\ I_{21} & I_{22} \end{pmatrix}, \qquad I^{-1}(\boldsymbol{\beta}) = \begin{pmatrix} I^{11}(\boldsymbol{\beta}) & I^{12}(\boldsymbol{\beta}) \\ I^{21}(\boldsymbol{\beta}) & I^{22}(\boldsymbol{\beta}) \end{pmatrix},$$

where I_{11}, I_{22}, I_{12} are the second (mixed) derivatives of $-LL$ in $\boldsymbol{\beta}_1$, $\boldsymbol{\beta}_2$, and I^{11} is the upper-left $q \times q$ block of I^{-1} (it involves both $\boldsymbol{\beta}_1$ and $\boldsymbol{\beta}_2$). Let $\hat{\boldsymbol{\beta}}_2(\mathbf{b})$ be the MLE of $\boldsymbol{\beta}_2$ under the constraint $\boldsymbol{\beta}_1 = \mathbf{b}$.

Definition 12.3: Local Wald, likelihood ratio and score tests

Each is approximately χ^2_q under H_0:

$$\chi^2_W = (\hat{\boldsymbol{\beta}}_1 - \mathbf{b})'[I^{11}(\hat{\boldsymbol{\beta}})]^{-1}(\hat{\boldsymbol{\beta}}_1 - \mathbf{b}), \qquad (\widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}}_1) = I^{11}(\hat{\boldsymbol{\beta}}))$$
$$\chi^2_{LR} = 2[LL(\hat{\boldsymbol{\beta}}) - LL(\mathbf{b}, \hat{\boldsymbol{\beta}}_2(\mathbf{b}))],$$
$$\chi^2_{SC} = \mathbf{U}_1(\mathbf{b}, \hat{\boldsymbol{\beta}}_2(\mathbf{b}))'\, I^{11}(\mathbf{b}, \hat{\boldsymbol{\beta}}_2(\mathbf{b}))\, \mathbf{U}_1(\mathbf{b}, \hat{\boldsymbol{\beta}}_2(\mathbf{b})),$$

$\mathbf{U}_1$ the $q \times 1$ score for $\boldsymbol{\beta}_1$.

In R: LR by fitting full and restricted models; Wald from the full-model estimates and covariance; score from the reduced-model MLEs used as an offset.

2.1 Linear combinations

Definition 12.4: Wald test of $H_0 : C\boldsymbol{\beta} = C\boldsymbol{\beta}_0$

For a $q \times p$ matrix C of full rank $q \le p$,

$$\chi^2_W = (C\hat{\boldsymbol{\beta}} - C\boldsymbol{\beta}_0)'[C\, I^{-1}(\hat{\boldsymbol{\beta}})\, C']^{-1}(C\hat{\boldsymbol{\beta}} - C\boldsymbol{\beta}_0) \;\approx\; \chi^2_q \quad \text{under } H_0.$$

Example 12.5: larynx cancer, stage adjusted for age

Model $h(t \mid \mathbf{Z}) = h_0(t)\exp\{\beta_1 Z_1 + \beta_2 Z_2 + \beta_3 Z_3 + \beta_4\,\text{age}\}$ (stage dummies as in Lecture 11).

- $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$ (no stage effect, age-adjusted): all three local tests are significant.
- $H_0 : \beta_1 = \beta_2 = \beta_3$, i.e. $\beta_1 - \beta_2 = 0$ and $\beta_2 - \beta_3 = 0$:

$$C = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \end{pmatrix}, \qquad \chi^2_W = \begin{pmatrix} \hat{\beta}_1 - \hat{\beta}_2 \\ \hat{\beta}_2 - \hat{\beta}_3 \end{pmatrix}' [C\,\widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}})\, C']^{-1} \begin{pmatrix} \hat{\beta}_1 - \hat{\beta}_2 \\ \hat{\beta}_2 - \hat{\beta}_3 \end{pmatrix} = 10.73,$$

  $p = P(\chi^2_2 \ge 10.73) = 0.0047$.

- **Linearity in stage**: $\beta_2 = 2\beta_1$, $\beta_3 = 3\beta_1$, i.e. $C = \begin{pmatrix} 2 & -1 & 0 & 0 \\ 3 & 0 & -1 & 0 \end{pmatrix}$; $p = P(\chi^2_2 \ge 2.47) = 0.29$: no evidence against a linear stage term.

3 Discretizing a Continuous Covariate

When: to ease interpretation, or when the log hazard ratio is not constant per unit increase. **How**: a biologically motivated cut point if available; otherwise a data-oriented one. The median split tends to perform poorly; the **outcome-oriented** approach picks the cut point C (e.g. $Z = I(\text{age} \ge C)$) giving the most significant two-group comparison.

Definition 12.6: Optimal cut point

For candidate cut points $C_1, C_2, \ldots$, let d_i^+, Y_i^+ be the failures and numbers at risk at t_i among subjects with covariate $\geq C_k$, and

$$S_k = \sum_{i=1}^{D} \left[d_i^+ - Y_i^+ \frac{d_i}{Y_i} \right].$$

$\hat{C}$ is the C_k maximizing $|S_k|$.

The usual two-sample test at $\hat{C}$ is **overly optimistic**: the cut point was chosen to maximize the separation.

Theorem 12.7: Contal and O'Quigley test

With

$$s^2 = \frac{1}{D-1} \sum_{i=1}^{D} \left\{ 1 - \sum_{j=1}^{i} \frac{1}{D-j+1} \right\}^2, \qquad Q = \frac{\max_k |S_k|}{s\sqrt{D-1}},$$

under H_0 the law of Q is approximately that of $\sup |B^0(x)|$, B^0 a Brownian bridge; for $Q > 1$, $p \approx 2\exp(-2Q^2)$.

4 Variable Selection

Criteria: Wald, LR and score tests, and

$$\text{AIC} = -2\log L + kp \qquad (k \text{ prespecified, e.g. } 2;\ p = \#\text{ regression parameters}),$$

smaller is better ("the simpler, the better").

1. Fix the candidate factors and the primary hypothesis.
2. Start with the hypothesized factors only.
3. Add the factor (possibly a set of covariates) with the smallest p-value given the current model.
4. Drop insignificant factors; repeat 3–4 until the fit no longer improves.

R: `My.stepwise.coxph` (package `My.stepwise`), appropriate when all factors are single covariates; check the final model for interpretability.

5 Estimating H_0 and Survival

5.1 Complete likelihood

Under the PH model,

$$L(\boldsymbol{\beta}, h_0) = \prod_{j=1}^{n} f(T_j)^{\delta_j} S(T_j)^{1-\delta_j} = \prod_{j=1}^{n} h(T_j)^{\delta_j} S(T_j) = \prod_{j=1}^{n} \left[h_0(T_j) e^{\boldsymbol{\beta}'\mathbf{Z}_j} \right]^{\delta_j} \exp\left\{ - H_0(T_j) e^{\boldsymbol{\beta}'\mathbf{Z}_j} \right\}.$$

Theorem 12.8: Breslow estimator

Obtain $\hat{\boldsymbol{\beta}}$ and $\widehat{V}(\hat{\boldsymbol{\beta}}) = I^{-1}(\hat{\boldsymbol{\beta}})$ from the partial likelihood, set $W(t_i;\hat{\boldsymbol{\beta}}) = \sum_{j\in R(t_i)} e^{\hat{\boldsymbol{\beta}}'\mathbf{Z}_j}$, and

$$\hat{H}_0(t) = \sum_{t_i \le t} \frac{d_i}{W(t_i;\hat{\boldsymbol{\beta}})},$$

which reduces to the Nelson–Aalen estimator (Theorem 5.5) without covariates. Then

$$\hat{S}_0(t) = \exp\{-\hat{H}_0(t)\}, \qquad \hat{S}(t \mid \mathbf{Z} = \mathbf{Z}_0) = \exp\{-\hat{H}_0(t)e^{\hat{\boldsymbol{\beta}}'\mathbf{Z}_0}\} = \hat{S}_0(t)^{\exp(\hat{\boldsymbol{\beta}}'\mathbf{Z}_0)},$$

asymptotically normal with variance $\widehat{V}[\hat{S}(t \mid \mathbf{Z}_0)]$ (K&M (8.8.5)–(8.8.8)).

5.2 Alternative estimators

With one event at each t_i:

- Kalbfleisch–Prentice (1973): $\tilde{H}_0(t) = \sum_{t_i \le t} \Big[1 - \Big(1 - \dfrac{\delta_i \exp(\hat{\boldsymbol{\beta}}'\mathbf{Z}_{(i)})}{W(t_i;\hat{\boldsymbol{\beta}})}\Big)^{\exp(-\hat{\boldsymbol{\beta}}'\mathbf{Z}_{(i)})}\Big]$.
- Product-integral (Kaplan–Meier type) versions of the Breslow and K–P estimators:

$$\hat{S}_2(t \mid \mathbf{0}) = \prod_{t_i \le t} \Big(1 - \frac{\delta_i}{W(t_i;\hat{\boldsymbol{\beta}})}\Big), \qquad \hat{S}_3(t \mid \mathbf{0}) = \prod_{t_i \le t} \Big(1 - \frac{\delta_i \exp(\hat{\boldsymbol{\beta}}'\mathbf{Z}_{(i)})}{W(t_i;\hat{\boldsymbol{\beta}})}\Big)^{\exp(-\hat{\boldsymbol{\beta}}'\mathbf{Z}_{(i)})},$$

  and $\hat{S}_k(t \mid \mathbf{Z}_0) = \hat{S}_k(t \mid \mathbf{0})^{\exp(\hat{\boldsymbol{\beta}}'\mathbf{Z}_0)}$, $k = 2, 3$.

All have similar asymptotic properties. Confidence intervals for $S(t \mid \mathbf{Z}_0)$: linear, log-log transformed (recommended), arcsine-square-root.

6 Selected exercises

Motivation. Breslow's tie-handling and Breslow's baseline estimator come from the same profile likelihood; the Contal–O'Quigley s^2 is a variance of log-rank scores.

Exercise 12.1 (Breslow's estimator as a profile likelihood)

In the complete likelihood let h_0 put mass $\lambda_i > 0$ at each t_i and none elsewhere, so $H_0(t) = \sum_{t_i \le t} \lambda_i$.
(a) For fixed $\boldsymbol{\beta}$, show that the maximizer is $\hat{\lambda}_i = d_i / W(t_i;\boldsymbol{\beta})$.
(b) Show that the profile likelihood $\max_\lambda L(\boldsymbol{\beta}, \lambda)$ is proportional to Breslow's $L_1(\boldsymbol{\beta})$ in (8.4.1).

Solution [Supplement].

(a) Subjects in $D(t_i)$ contribute $\lambda_i e^{\boldsymbol{\beta}'\mathbf{Z}_j}$, and each subject j contributes $\exp\{-e^{\boldsymbol{\beta}'\mathbf{Z}_j} \sum_{t_i \le T_j} \lambda_i\}$. Collecting by t_i (j is in $R(t_i)$ iff $t_i \le T_j$),

$$\log L = \sum_{i=1}^{D} \left[d_i \log \lambda_i + \boldsymbol{\beta}'\mathbf{S}_i - \lambda_i W(t_i;\boldsymbol{\beta})\right],$$

and $\partial/\partial\lambda_i = d_i/\lambda_i - W(t_i;\boldsymbol{\beta}) = 0$; the function is concave in λ_i.

(b) Substituting,

$$\log L(\boldsymbol{\beta}, \hat{\lambda}) = \sum_{i=1}^{D} \left[\boldsymbol{\beta}' \mathbf{S}_i - d_i \log W(t_i; \boldsymbol{\beta})\right] + \sum_{i=1}^{D} (d_i \log d_i - d_i) = \log L_1(\boldsymbol{\beta}) + \text{const.}$$

So $\hat{H}_0$ in Theorem 12.8 is $\sum_{t_i \le t} \hat{\lambda}_i$ at $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}$. ■

Exercise 12.2 (The Contal–O'Quigley scores)

Let $a_i = 1 - \sum_{j=1}^{i} \frac{1}{D-j+1}$, $i = 1, \ldots, D$.
(a) Show that $\sum_{i=1}^{D} a_i = 0$, so s^2 is the sample variance of the a_i.
(b) Show that with no censoring and no ties, $a_i = \delta_i - \hat{H}(t_i)$, where $\hat{H}$ is the Nelson–Aalen estimator of the pooled sample.

Solution [Supplement].

(a) Swap the sums: $\sum_{i=1}^{D} \sum_{j=1}^{i} \frac{1}{D-j+1} = \sum_{j=1}^{D} \frac{D-j+1}{D-j+1} = D$, so $\sum_i a_i = D - D = 0$ and $s^2 = \frac{1}{D-1} \sum_i (a_i - \bar{a})^2$.

(b) Without censoring the jth death has $D - j + 1$ at risk, so $\hat{H}(t_i) = \sum_{j \le i} \frac{1}{D-j+1}$ and $\delta_i = 1$. ■